

How to login to Your System ?

1. Locally.
2. Remotely.

Locally: through the System interface and using System Keyboard, there is two types of login:

- a. GUI: through Graphical interface
- b. TUI: through the multi-user or Text User interface where's no Graphical.
 - To switch between GUI and TUI you can use **(alt + ctrl + f(1-6))** or **chvt** command like the following :
[root@local]# chvt 6 (where 6 is the tty number)

TTY Vs PTS ?

Tty is the name of shell from TUI mode .

Pts is the name of terminal from GUI or remote login.

You need to know the difference between tty and pts for this:

who will show all users who has a login-tty

```
$ who
Root pts/0      2020-22-09 11:33
adel pts/1      2020-22-09 11:45
```

w will Show who is logged on and what they are doing, so it is like **who**, plus the "what they are doing" info:

```
$ w
12:12:01 up 78 days, 15:52,  2 users,  load average: 0.25, 0.34, 0.29
USER TTY      FROM          LOGIN@  IDLE   JCPU   PCPU   WHAT
bo   pts/0      67.xx.xx.xx   11:33   0.00s  0.13s  0.00s  sshd: bo[priv]
bo   pts/1      67.xx.xx.xx   11:45   26:49  0.04s  0.04s  -bash
```

whoami

prints the logged on user

```
$ whoami
bo
```

PWD

Print working directory, what's the path you already in.

```
$ pwd
/home/adel
```

\$ date >> prints the date and time

Basic Shell Skills:

1. **Command syntax:**
command -options arguments

LS command for example:

ls—List directory contents.

Examples

```
[me@linuxbox ~]$ ls
Desktop Documents Music Pictures Public Templates Videos
```

```
me@linuxbox ~]$ ls /usr
bin games kerberos libexec/sbin src
etc include lib local share tmp
```

```
[me@linuxbox ~]$ ls -l
total 56
drwxrwxr-x 2 me me 4096 2012-10-26 17:20 Desktop
drwxrwxr-x 2 me me 4096 2012-10-26 17:20 Documents
drwxrwxr-x 2 me me 4096 2012-10-26 17:20 Music
drwxrwxr-x 2 me me 4096 2012-10-26 17:20 Pictures
drwxrwxr-x 2 me me 4096 2012-10-26 17:20 Public
drwxrwxr-x 2 me me 4096 2012-10-26 17:20 Templates
drwxrwxr-x 2 me me 4096 2012-10-26 17:20 Videos
```

ls syntax

\$ ls [*options*] [*file|dir*]

ls command options

ls command main options:

option	description
ls -a	list all files including hidden file starting with '.'
ls --color	colored list [=always/never/auto]
ls -d	list directories - with ' */'
ls -F	add one char of */=>@ to enteries
ls -i	list file's inode index number
ls -l	list with long format - show permissions
ls -la	list long format including hidden files
ls -lh	list long format with readable file size
ls -ls	list with long format with file size
ls -r	list in reverse order
ls -R	list recursively directory tree

[`ls -s`](#) list file size
[`ls -S`](#) sort by file size
[`ls -t`](#) sort by time & date
`ls -X` sort by extension name

ls command examples

You can press the *tab* button to auto complete the file or folder names.

List directory *Documents/Books* with *relative* path:

```
$ ls Documents/Books
```

List directory */home/user/Documents/Books* with *absolute* path.

```
$ ls /home/user/Documents/Books
```

List root directory:

```
$ ls /
```

List parent directory:

```
$ ls ..
```

List user's home directory (e.g: */home/user*):

```
$ ls ~
```

List with long format:

```
$ ls -l
```

Show hidden files:

```
$ ls -a
```

List with long format and show hidden files:

\$ ls -la

Sort by date/time:

\$ ls -t

Sort by file size:

\$ ls -S

List all subdirectories:

\$ ls *

Recursive directory tree list:

\$ ls -R

2. Search for Certain a specific Command:

\$mandb

\$man -k [KeyWord]

3. Execute Multiple Commans in one line

\$ ls -l / ; firewall-cmd --reload

4. Type your Command in Multiple lines:

```
# nmcli con mod en192 \  
>ipv4.addresses 192.168.5.240/24 \  
>ipv4.gateway 192.168.4.60 \  
>ipv4.dns 8.8.8.8
```